

Qinfeng Zhu

Ph.D. Candidate in Computer Science, University of Liverpool

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Seeking 2026 research internships and 2027 full-time research roles.

Education

University of Liverpool

Expected 2027

Ph.D. in Computer Science, advised by Dr. Lei Fan and Dr. Anh Nguyen

University of Liverpool

2023

Master of Research in Computer Science, advised by Dr. Ming Xu

Research Experience

1. Embodied Intelligence: Agentic Harness (*Ongoing*)

- Exploring the use of large language models (LLMs) and vision-language-action (VLA) models within agentic harness architectures for embodied intelligence.

2. Panoramic and Spherical Vision

- **Sphere-native perception:** Study intrinsic spherical features for rotation-robust panoramic dense prediction, alongside a survey organizing scene-understanding methods from distortion-aware engineering to sphere-native modeling. [2, 3]
- **Full-sphere tracking:** Investigate person tracking from both fixed and moving panoramic cameras, extending the current research focus from dense scene prediction to dynamic full-sphere perception. [1]

3. 3D Scene Understanding and Multispectral Perception

- **Point-cloud learning:** Authored a survey of data augmentation for point-cloud deep learning and contributed co-first-author work on the SemanticUrban dataset for urban 3D semantic understanding. [6, 4]
- **Multispectral scene understanding:** Study semantic segmentation with multispectral imagery in indoor and environmental settings, including datasets for indoor scenes and lakeside-waste monitoring. [9, 10]

4. Remote Sensing and Earth Observation

- **Segmentation methods:** Published work addresses category-specific fusion, Vision Mamba scanning strategies, hybrid local-global modeling, and pre-training for semantic segmentation of remotely sensed imagery. [5, 8, 11, 7]
- **Geospatial risk modeling:** Ongoing work investigates satellite embeddings and landslide conditioning factors for deep-learning-based landslide susceptibility mapping. [12]

Academic Service

Conference reviewer: ICLR, AAAI, IJCAI, NeurIPS, WACV.

Journal reviewer: TNNLS, TCSVT, T-ITS, TNSE, ISPRS JPRS, PR, NN, ESWA, EAAI, Neurocomputing.

Selected Publications

[1] PanoPed: Tracking People on the Full Sphere from Fixed and Moving Cameras

Qinfeng Zhu, Weiguang Zhao, Yunxi Jiang, Anh Nguyen, Lei Fan.

Under review at ICLR 2027.

[2] SO3UFormer: Learning Intrinsic Spherical Features for Rotation-Robust Panoramic Dense Prediction

Qinfeng Zhu, Yunxi Jiang, Lei Fan.

arXiv, 2026. Under review at International Journal of Computer Vision (IJCV).

[3] Panoramic Scene Understanding: A Survey from Distortion-Aware Engineering to Sphere-Native Modeling

Qinfeng Zhu, Lei Fan.

arXiv, 2026. Under review at International Journal of Computer Vision (IJCV).

- [4] **Advancements in Point Cloud Data Augmentation for Deep Learning: A Survey**
Qinfeng Zhu, Lei Fan, Ningxin Weng.
Pattern Recognition (PR), 2024.
- [5] **ClassWise-CRF: Category-Specific Fusion for Enhanced Semantic Segmentation of Remote Sensing Imagery**
Qinfeng Zhu, Yunxi Jiang, Lei Fan.
Neural Networks (NN), 2026.
- [6] **Towards Accurate Urban Scene Understanding Using Point Clouds: The SemanticUrban Dataset**
Yuan Fang, **Qinfeng Zhu** (equal), Yuanzhi Cai, Lei Fan.
Expert Systems with Applications (ESWA), 2026.
- [7] **Rethinking Scanning Strategies with Vision Mamba in Semantic Segmentation of Remote Sensing Imagery**
Qinfeng Zhu, Yuan Fang, Yuanzhi Cai, Cheng Chen, Lei Fan.
IEEE JSTARS, 2024.
- [8] **SwinMamba: A Hybrid Local-Global Mamba Framework for Enhancing Semantic Segmentation of Remotely Sensed Images**
Qinfeng Zhu, Han Li, Liang He, Lei Fan.
Digital Signal Processing, 2026.
- [9] **IndoorMS: A Multispectral Dataset for Semantic Segmentation in Indoor Scene Understanding**
Qinfeng Zhu, Jingjing Xiao, Lei Fan.
IEEE Sensors Journal, 2025.
- [10] **Enhancing Environmental Monitoring Through Multispectral Imaging: The WasteMS Dataset for Semantic Segmentation of Lakeside Waste**
Qinfeng Zhu, Ningxin Weng, Lei Fan, Yuanzhi Cai.
MMM, 2025.
- [11] **A Generalised Pre-Training Strategy for Deep Learning Networks in Semantic Segmentation of Remotely Sensed Images**
Yuan Fang, Yuanzhi Cai, Jagannath Aryal, **Qinfeng Zhu**, Hong Huang, Cheng Zhang, Lei Fan.
CGI / LNCS, 2026.
- [12] **From Landslide Conditioning Factors to Satellite Embeddings: Evaluating the Utilisation of Google AlphaEarth for Landslide Susceptibility Mapping Using Deep Learning**
Yusen Cheng, **Qinfeng Zhu**, Lei Fan.
arXiv, 2026. *Under review*.